

# Mosaovir Ahmed Khan

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## SUMMARY

Electrical & Computer Engineering candidate (EIT-eligible 2026) with hands-on experience in SCADA modernization, PLC/RTU validation, leak detection systems, and midstream OT infrastructure. Experienced in CAT/SAT testing, control logic verification, signal commissioning, and HMI deployment in live industrial environments. Strong understanding of production-critical systems, regulatory compliance (CEC/CSA), and cross-functional coordination between Engineering, Control Center, and Commercial Operations. Prepared for field commissioning, reliability engineering, and OT support roles in upstream, midstream, and downstream operations.

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## EDUCATION

### University of Calgary

Expected April 2026

Bachelor of Science in Electrical and Computer Engineering

- Dean's Honor Roll 2021-2026 (3.7 GPA)
- President's Admission Scholarship Recipient, Continuing Undergraduate Student Scholarship Recipient

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## AREAS OF EXPERTISE

**Industrial Automation & OT:** SCADA Architecture, PLC/RTU Validation, RSLogix 5000, HMI (L1/L2), Leak Detection Systems

**Programming & Data Automation:** Python, SQL, C/C++, C#, VBA, MATLAB/Simulink

**Electrical & Control Systems:** Allen-Bradley PLCs, Flow Computers, VFDs, Power Electronics, Motor Drives, Analog/Digital Circuits, Embedded Systems

**Standards & Safety:** Canadian Electrical Code (CEC), CSA, IEEE, Alberta & Saskatchewan Industrial Codes, H2S Alive, First Aid, WHMIS, OHS, Basic Firefighting Training

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## PROFESSIONAL EXPERIENCE

### Cenovus Energy

September 2024 – August 2025

Midstream OT Systems & Infrastructure | Co-Op

- Developed and implemented the OCM (Organizational Change Management) framework for the SCADA Upgrade project, aligning operator workflows, HMI updates, and documentation to support rollout and minimize downtime.
- Supported FAT/SAT-style testing for SCADA, PLC, and leak detection systems, including signal validation, logic verification, and documentation of test results.
- Validated over ten thousand I/O points, alarm tags, and control signals prior to deployment.
- Interpreted and updated single-line diagrams and control schematics to update and improve L1 and L2 HMI graphics.
- Worked with integrator teams to review panel wiring documentation, device configurations, and SCADA integration requirements.
- Streamlined workflow using automation with SQL, C++, Python and VBA and saving hundreds of hours of manual work.
- Participated in field site visits and panel inspections, supporting safe maintenance activities under Cenovus safety protocols and Life Saving Rules in live industrial environments.
- Collaborated across multiple teams such as Engineering, Measurement, Commercial Operations and Control Center to ensure alignment between OT Systems and production requirements.

### University of Calgary

May 2024 – August 2024

Intelligent Dynamics & Control Laboratory | Research Internship

- Used C++ and Python to program onboard computers and collaborated on multiple projects at the Intelligent Dynamics and Control Lab (IDCL).
- Designed the communication protocol for the Dynamic Docking System of a bi copter.
- Developed a dynamic controller for a UGV to work in tandem with the bi copter.

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## PROJECTS

### Capstone Power Architecture Design

September 2025 – April 2025

Project Lead | University of Calgary

- Designed regulated multi-rail power distribution architecture incorporating switching converters, EMI mitigation, thermal management, and fault isolation.
- Developed a custom PCB for regulated power delivery, incorporating high frequency switching converters, filtering networks, and thermal/EMI considerations.
- Implemented comprehensive system protection to safeguard high-value sensors and computer hardware.

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- Conducted load analysis, ripple analysis, efficiency evaluation, and component derating to ensure system reliability under real-time dynamic load conditions.
- Produced wiring diagrams, schematics, and documentation to support integration, testing, and commissioning during hardware-in-the-loop trials.

## Grid Connected PV System Design

September 2025 – December 2025

Course Project | University of Calgary

- Designed a complete grid-tied solar PV system for a residential load using industry-grade simulation software (PV Syst).
- Performed detailed power flow, fault level, and grid-stability analysis, including impedance matching, voltage regulation compliance, and utility back-feed limits.
- Modeled DC/DC and DC/AC converter stages, evaluated inverter control behavior, and optimized system performance based on converter efficiency curves, switching losses, and thermal constraints.
- Conducted sizing and selection of PV modules, string configurations, wiring, protection devices, grounding, and isolation hardware to meet CEC/CSA standards.
- Generated single-line diagrams, loss calculations, and performance forecasts to validate the system's reliability, safety, and annual energy yield.

## Real Time Embedded Control System

September 2025 – December 2025

Course Project | University of Calgary

- Designed and implemented a real-time embedded timer/stopwatch system using Free RTOS on a PIC 24 microcontroller.
- Built to work with hard real time constraints to maintain 99.99% accuracy and extremely low latencies.
- Developed multi-task architecture (timekeeping, display refresh, input handling) with priority-based scheduling to maintain sub-millisecond timing accuracy (<1 ms jitter).
- Configured hardware interrupts and timers to ensure bounded response time under concurrent task load.
- Measured and optimized CPU utilization, reducing task latency by 30% through scheduling and ISR optimization.
- Implemented watchdog logic and fault recovery routines to simulate industrial fail-safe mechanisms.
- Validated real-time performance using oscilloscope timing measurements and logic analysis.

## EXTRA-CURRICULAR INVOLVEMENT

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### Engineering Leadership Program

January 2022 – April 2025

University of Calgary

- Progressive Skill Development: Begins with foundational leadership skills, advancing to team leadership concepts.
- Practical Application: Blends theoretical knowledge with real-world leadership scenarios in engineering.

### Community Involvement

September 2024 – August 2025

Cenvous Cares

- Volunteered at local schools to cook lunch for 200 students.
- Involved with local motor vehicle enthusiasts club and helped organize events and fundraisers.
- Involved with local indigenous communities and helped organize cultural events and fundraisers.